ENTERPRISE NETWORK INFRASTRUCTURE

DESIGN, CONFIGURATION & VALIDATION REPORT

Prepared as a simulated network implementation for a fictional company

Report Item	Details
Environment	Cisco Packet Tracer simulation
Architecture	Core–Access switching with two routers
Network Services	VLANs, VTP, 802.1Q trunks, ROAS, DHCP, HSRP, PVST, PortFast, and BPDU Guard
Endpoint Scope	Six client PCs and one dedicated DHCP server
Status	Configuration and validation report; actual addressing/interface values should be inserted from the Packet Tracer file

Project Scope Note: This report describes the topology and technologies reported as implemented. The VLAN networks and server address in this report are based on the project details provided. Router interface addresses and HSRP virtual gateways are documented as a consistent addressing convention and should be checked against the final running configuration.

1. EXECUTIVE SUMMARY

This project models the network infrastructure of a fictional company with multiple user groups. The design uses a central multilayer switch, two access switches, two routers, six PCs, and a dedicated DHCP server. The network is segmented into VLANs 10, 20, 30, and 40. VLANs 10 and 20 receive dynamic addressing from router-based DHCP pools, while the server in VLAN 40 provides DHCP service for clients in VLAN 30. Router-on-a-Stick (ROAS) is used for inter-VLAN routing, and HSRP provides a shared virtual default gateway across the two routers. VTP, 802.1Q trunking, PVST, PortFast, and BPDU Guard support VLAN administration, loop prevention, and safer access-port operation.

2. BUSINESS SCENARIO AND OBJECTIVES

The fictional company requires a manageable campus LAN that separates departments logically, reduces manual endpoint configuration, and continues to offer a default-gateway service if the active HSRP router becomes unavailable. The network is designed as a small campus-style environment with a central switching point and two access-layer switches serving end users.

- Separate client groups using VLANs rather than placing all endpoints in one broadcast domain.
- Distribute VLAN information consistently across participating switches using VTP.
- Carry multiple VLANs over configured trunk links using IEEE 802.1Q.
- Provide inter-VLAN communication through Router-on-a-Stick subinterfaces.
- Automate IP address allocation using router DHCP pools and a dedicated DHCP server.
- Provide a shared virtual default gateway using HSRP.
- Prevent Layer 2 switching loops using PVST and improve edge-port behavior with PortFast and BPDU Guard.
- Validate connectivity and use show commands and ping tests to support troubleshooting.

3. PHYSICAL AND LOGICAL TOPOLOGY

The simulated topology contains the following devices and relationships:

- **Core Layer:** One central multilayer switch acting as the core switching point.
- **Access Layer:** Two access switches, each connected to three client PCs.
- **Routing & Redundancy:** Two routers connected to the core switch; configured for HSRP and Router-on-a-Stick.
- **Services:** One server connected to an access port on the core switch and assigned to VLAN 40.
- **Layer 2 Links:** Inter-switch links between core and access switches, plus a link between the two access switches. These redundant paths make STP behavior relevant.

Logical traffic separation is based on VLAN membership. Each access switch has endpoints associated with VLANs 10, 20, and 30, while the DHCP server is placed in VLAN 40. Exact physical interface numbers should be documented from the final Packet Tracer file.

4. VLAN AND ADDRESSING PLAN

VLAN	Purpose	Connected Devices / Service	Address Allocation
10	Client Network A	PC clients on both access switches	192.168.10.0/24; HSRP GW 192.168.10.1; Router IPs .2 and .3
20	Client Network B	PC clients on both access switches	192.168.20.0/24; HSRP GW 192.168.20.1; Router IPs .2 and .3
30	Client Network C	PC clients on both access switches	192.168.30.0/24; HSRP GW 192.168.30.1; DHCP via Server relay
40	Server Network	Dedicated DHCP server	192.168.40.0/24; Server IP 192.168.40.100; HSRP GW 192.168.40.1

The addressing plan uses /24 networks (255.255.255.0). The HSRP virtual gateway is set to 192.168.X.1 for each VLAN, with router physical addresses 192.168.X.2 and 192.168.X.3 as a consistent convention.

5. NETWORK TECHNOLOGIES AND IMPLEMENTATION

5.1 VLAN Segmentation

VLANs 10, 20, and 30 separate user groups into distinct Layer 2 broadcast domains. VLAN 40 isolates the DHCP server network, improving logical separation and simplify troubleshooting.

5.2 VTP (VLAN Trunking Protocol)

The core switch is configured as the VTP server, while access switches operate as VTP clients. This allows VLAN database information to be advertised across trunk links. VTP domain name, version, and password must match across switches.

5.3 802.1Q Trunking

Trunk links carry traffic for multiple VLANs between switches and routers. Allowed VLAN lists and native VLAN settings must be consistent at both ends. End-device ports remain access ports.

5.4 Router-on-a-Stick (ROAS)

Each router uses logical subinterfaces on its physical Ethernet connection with 802.1Q encapsulation. This enables inter-VLAN routing through the routers in coordination with HSRP.

5.5 HSRP (Hot Standby Router Protocol)

HSRP provides a virtual IP address used as default gateway. One router is active and forwards traffic; the standby takes over if the primary fails, ensuring seamless default gateway redundancy.

5.6 DHCP Services

Router-based DHCP pools supply addresses to VLANs 10 and 20. The server in VLAN 40 supplies addresses to VLAN 30 via `ip helper-address 192.168.40.100` on the VLAN 30 router subinterface. Router pools exclude gateway and reserved addresses.

5.7 PVST / STP

Per-VLAN Spanning Tree (PVST) runs a separate STP instance for each VLAN to prevent Layer 2 loops while maintaining optimal path selection across redundant inter-switch links.

5.8 PortFast and BPDU Guard

PortFast allows edge ports to enter forwarding state immediately. BPDU Guard protects against rogue switch connections by error-disabling edge ports if a BPDU is received.

6. CONFIGURATION REFERENCE (ILLUSTRATIVE)

6.1 VLANs and VTP Configuration

```
! Core switch (VTP server)
vtp mode server
vlan 10 | name CLIENT_A
vlan 20 | name CLIENT_B
vlan 30 | name CLIENT_C
vlan 40 | name SERVER

! Access switches (VTP clients)
vtp mode client
```

6.2 Access Ports and Edge Protection

```
interface <PC-facing-interface>
 switchport mode access
 switchport access vlan <10-or-20-or-30>
 spanning-tree portfast
 spanning-tree bpduguard enable

interface <server-facing-interface>
 switchport mode access
 switchport access vlan 40
```

6.3 Trunk Links Configuration

```
interface <switch-uplink>
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40
 ! Configure matching native VLAN at both ends
```

6.4 Router-on-a-Stick and HSRP

```
interface <router-interface>.10
 encapsulation dot1Q 10
 ip address 192.168.10.2 255.255.255.0
 standby 10 ip 192.168.10.1

interface <router-interface>.30
 encapsulation dot1Q 30
 ip address 192.168.30.2 255.255.255.0
 standby 30 ip 192.168.30.1
 ip helper-address 192.168.40.100
```

6.5 Router DHCP Pool Configuration

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

7. VALIDATION AND TROUBLESHOOTING PLAN

Area	Command / Test	Expected Evidence
VLAN Database	`show vlan brief`	VLANs 10, 20, 30, 40 exist and access ports are assigned correctly.
VTP Status	`show vtp status`	Correct VTP mode, domain, and database propagation verified.
Trunking	`show interfaces trunk`	Required VLANs allowed and active on intended trunk links.
STP Topology	`show spanning-tree vlan X`	Root bridge and blocking/alternate ports identified.
Port Protection	`show spanning-tree detail`	Edge ports have PortFast/BPDU Guard enabled; uplinks clean.
HSRP Redundancy	`show standby brief`	One Active and one Standby router per configured VLAN group.
DHCP Relay	`ipconfig /all` on PCs	Correct IP, mask, gateway, and DNS assigned to clients.
Failover Test	Shutdown active router interface	Standby router takes over; ping traffic resumes automatically.

8. ACCEPTANCE CRITERIA

- Each PC receives an IP address from its corresponding VLAN DHCP scope.
- Each client uses the HSRP virtual IP as its default gateway.
- VLAN 30 DHCP requests successfully traverse the router relay to VLAN 40 server.
- PortFast and BPDU Guard are restricted strictly to edge device access ports.

9. SECURITY & PRODUCTION CONSIDERATIONS

For production deployment, implement SSH, strong authentication, DHCP Snooping, Dynamic ARP Inspection (DAI), and Port Security to prevent unauthorized access and spoofing attacks.

APPENDIX A — VALUES TO RECORD FROM PACKET TRACER FILE

Item Description	Recorded Field / Parameter
Company / Scenario Name	_____
VTP Domain / Version / Password	_____
VLAN 10 Subnet & Gateway	192.168.10.0/24 | GW: 192.168.10.1
VLAN 20 Subnet & Gateway	192.168.20.0/24 | GW: 192.168.20.1
VLAN 30 Subnet & Gateway	192.168.30.0/24 | GW: 192.168.30.1
VLAN 40 Server IP Address	192.168.40.100 /24
Router Subinterface Mapping	_____
STP Root Bridge & Blocked Ports	_____